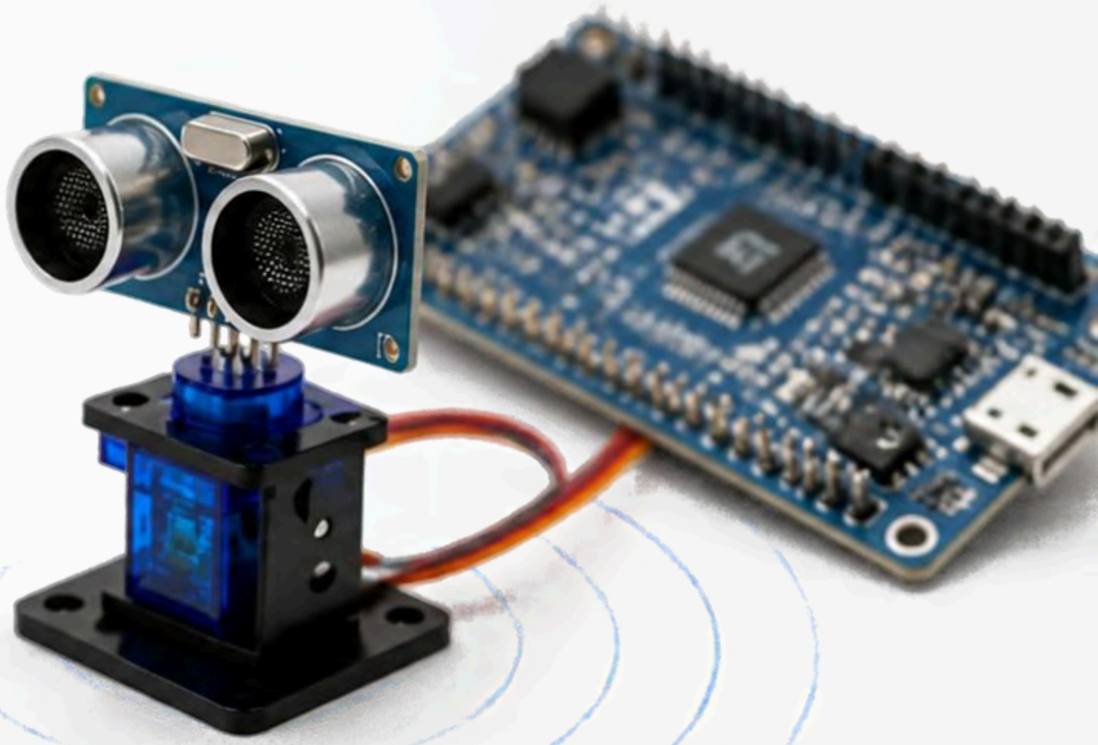
STM32-BASED RADAR TARGET DETECTION & THREAT ALERT SYSTEM

An ultrasonic scanning prototype with an adjustable safety threshold, local alerts and Bluetooth reporting.



ULTRASONIC | SERVO SCAN | ALERT | BLUETOOTH

Radar-inspired presentation; sensing is ultrasonic.
Servo angle provides approximate direction—not true radar tracking.

PROBLEM STATEMENT

Detect a nearby object and warn the operator immediately—without the cost and complexity of conventional radar.



1 SENSE
measure the nearest obstacle using the ultrasonic echo.

2 SCAN
move the sensor with the required SG90 servo.

3 DECIDE
compare distance with the live user-set threshold.

4 ALERT
trigger local outputs to warn the operator.

5 REPORT
send system status updates via Bluetooth.

LOW COST | **MODULAR** | **REAL-TIME** | **DEMONSTRABLE**

MOTIVATION

01 MOTIVATION

Easy to demonstrate: a live perimeter-style alert is visual, audible and immediately understandable.

Strong embedded depth: timers, ADC, GPIO, I°C, UART and PWM on one STM32 platform.

Affordable and repeatable: widely available modules keep the build practical for a student team.

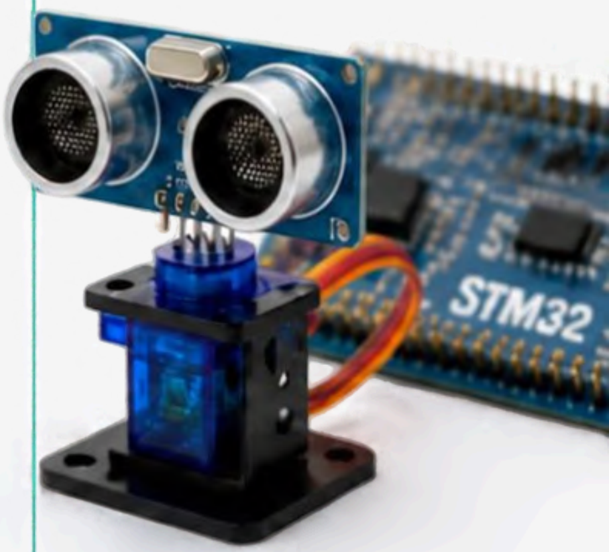
02 NOVELTY AND INNOVATION

Adjustable threshold: change the alert boundary through ADC input without re-flashing.

Multi-channel alerts: LCD, LEDs, buzzer and Bluetooth communicate one system state.

Servo-based sweep: pair each distance sample with an approximate scan angle.

Clearly defined scope: ultrasonic distance only; no speed, identity or multi-target claim.



OBJECTIVES AND DELIVERABLES

OUR OBJECTIVES

1 MEASURE DISTANCE
Use STM32 timer capture for the ultrasonic echo.

2 SET THE THRESHOLD
Read the potentiometer through the ADC.

3 SHOW AND ALERT
Drive the LCD, LEDs and buzzer from one state.

4 SCAN AND REPORT
Operate the servo and send status via Bluetooth.



OUR DELIVERABLES

WORKING PROTOTYPE
A complete integrated hardware system with all required modules.

FIRMWARE + DOCUMENTATION
STM32 source code, block diagram and final wiring documentation.

RECORDED TEST EVIDENCE
Measured results, final report and a repeatable live demonstration.

PROTOTYPE | FIRMWARE | WIRING | TEST DATA | REPORT | DEMO

HARDWARE SPECIFICATIONS

ALL COMPONENTS ARE MANDATORY



STM32F401CCU6
Main controller
84 MHz Cortex-M4



HC-SR04 compatible
Ultrasonic sensing
2–400 cm nominal



SG90 micro servo
Scanning motion
PWM controlled



Green + red LEDs
Visual status
With resistors



Push button
Mode / mute
Internal pull-up



HC-05 Bluetooth
Wireless reporting
UART serial



16×2 I²C LCD
Operator display
Distance + status



10 kΩ potentiometer
Threshold setting
Analog input



Active buzzer
Audible warning
GPIO controlled



Build essentials
Power + protection
5 V, breadboard, wires, ECHO divider

Final pin assignments and ratings must be verified against the selected module datasheets before wiring.

SYSTEM ARCHITECTURE

INPUTS



HC-SR04 SENSOR

TRIG + ECHO
Timer capture



POTENTIOMETER

ADC
Adjustable threshold



PUSH BUTTON

Mode, mute
or acknowledge

STM32F401CCU6

84 MHz Cortex-M4

- TIMER
- ADC
- GPIO
- I²C
- UART
- PWM

PROTECT STM32

Reduce 5 V ECHO to 3.3 V

OUTPUTS



LOCAL INTERFACE

16×2 LCD
Green/Red LEDs
Buzzer



SG90 SERVO

PWM sweep
for approximate
direction



HC-05 BLUETOOTH

UART
Distance, angle
and status reporting

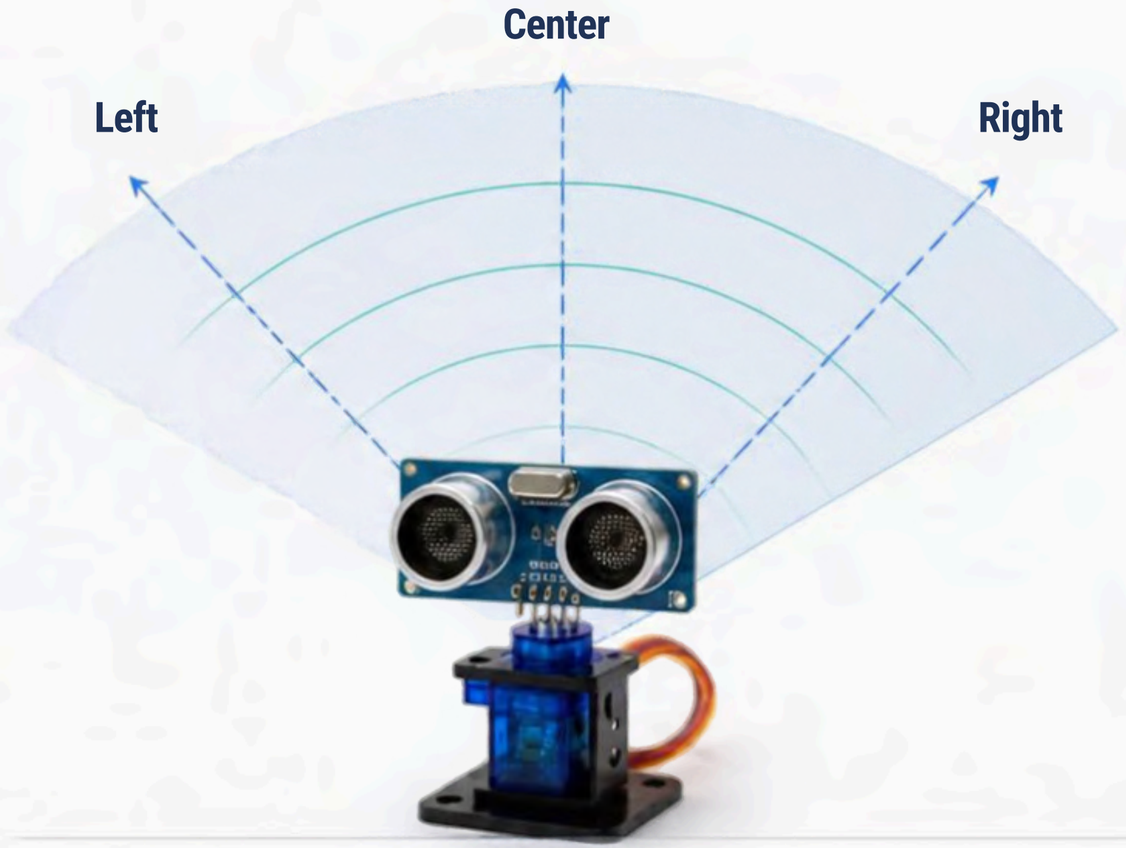
ULTRASONIC SENSING CAPABILITIES

2–400 cm
Nominal sensing range
Suitable for a compact indoor demonstration.

40 kHz
Ultrasonic carrier
Operates above the range of human hearing.

Real time
Fast distance updates
Supports immediate threshold-based warnings.

Non-contact
Safe object detection
No physical contact is required to sense an obstacle.



TIME-OF-FLIGHT PRINCIPLE
Distance = (echo time × speed of sound) ÷ 2.

SERVO-SCAN ADVANTAGE
The SG90 adds an approximate direction to every distance sample, creating a simple sector scan.

DESIGN CONSIDERATIONS
Keep targets within range; account for soft or angled surfaces, temperature and the nearfield blind zone.

LOW COST

SIMPLE INTERFACE

REPEATABLE MEASUREMENTS

IDEAL FOR SHORT-RANGE ALERTING

CONCLUSION AND IMPLEMENTATION PLAN



01
HARDWARE
safe wiring and sensor validation.

02
DECISION LOGIC
ADC threshold and filtered state.

03
SCANNING
servo sweep and angle calibration.

04
LOCAL ALERTS
LCD, LEDs, and buzzer indications.

05
BLUETOOTH
distance, angle, and state text reporting.

06
PROOF
test data, report, and live demo.

READY TO BUILD

A complete, affordable STM32 prototype with scanning, local alerts and Bluetooth reporting.

DELIVERABLES: | **PROTOTYPE** | **FIRMWARE** | **WIRING DIAGRAM** | **RECORDED MEASUREMENTS** | **REPORT** | **DEMONSTRATION**